

Dietary Epidemiology of Essential Tremor: Meat Consumption and Meat Cooking Practices

Background/Aim Harmane [1-methyl-9H-pyrido(3,4-b)indole] is a tremor-producing neurotoxin. Blood harmane concentrations are elevated in essential tremor (ET) patients for unclear reasons. Potential mechanisms include increased dietary harmane intake (especially through well-cooked meat) or genetic-metabolic factors. We tested the hypothesis that meat consumption and level of meat doneness are higher in ET cases than in controls. **Methods** Detailed data were collected using the Lawrence Livermore National Laboratory Meat Questionnaire. **Results** Total current meat consumption was greater in men with than without ET (135.3 ± 71.1 vs. 110.6 ± 80.4 g/day, $p = 0.03$) but not in women with versus without ET (80.6 ± 50.0 vs. 79.3 ± 51.0 g/day, $p = 0.76$). In an adjusted logistic regression analysis in males, higher total current meat consumption was associated with ET (OR = 1.006, $p = 0.04$, i.e., with 10 additional g/day of meat, odds of ET increased by 6%). Male cases had higher odds of being in the highest than lowest quartile of total current meat consumption (adjusted OR = 21.36, $p = 0.001$). Meat doneness level was similar in cases and controls. **Conclusion** This study provides evidence of a dietary difference between male ET cases and male controls. The etiological ramifications of these results warrant additional investigation. Copyright © 2008 S. Karger AG, Basel